

Mount Kenya



University

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STUDENT REPORTING LOGBOOK

BIT3208: Advanced Web Design and Development

Student Details

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Unit Name: Advanced Web Design and Development

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Project Details

Project Title: Task Manager System

Selected Technologies: PHP + MySQL

GitHub Repository Link: <https://github.com/Martin254-gif/task-manager>

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WEEK 1: LOCAL ENVIRONMENT SETUP

TITLE: Installation and testing of local Development Environment

FIG 1;Successful installation of Xampp on mac Os



Fig 2;Apache and MySQL services started in XAMPP control pannel

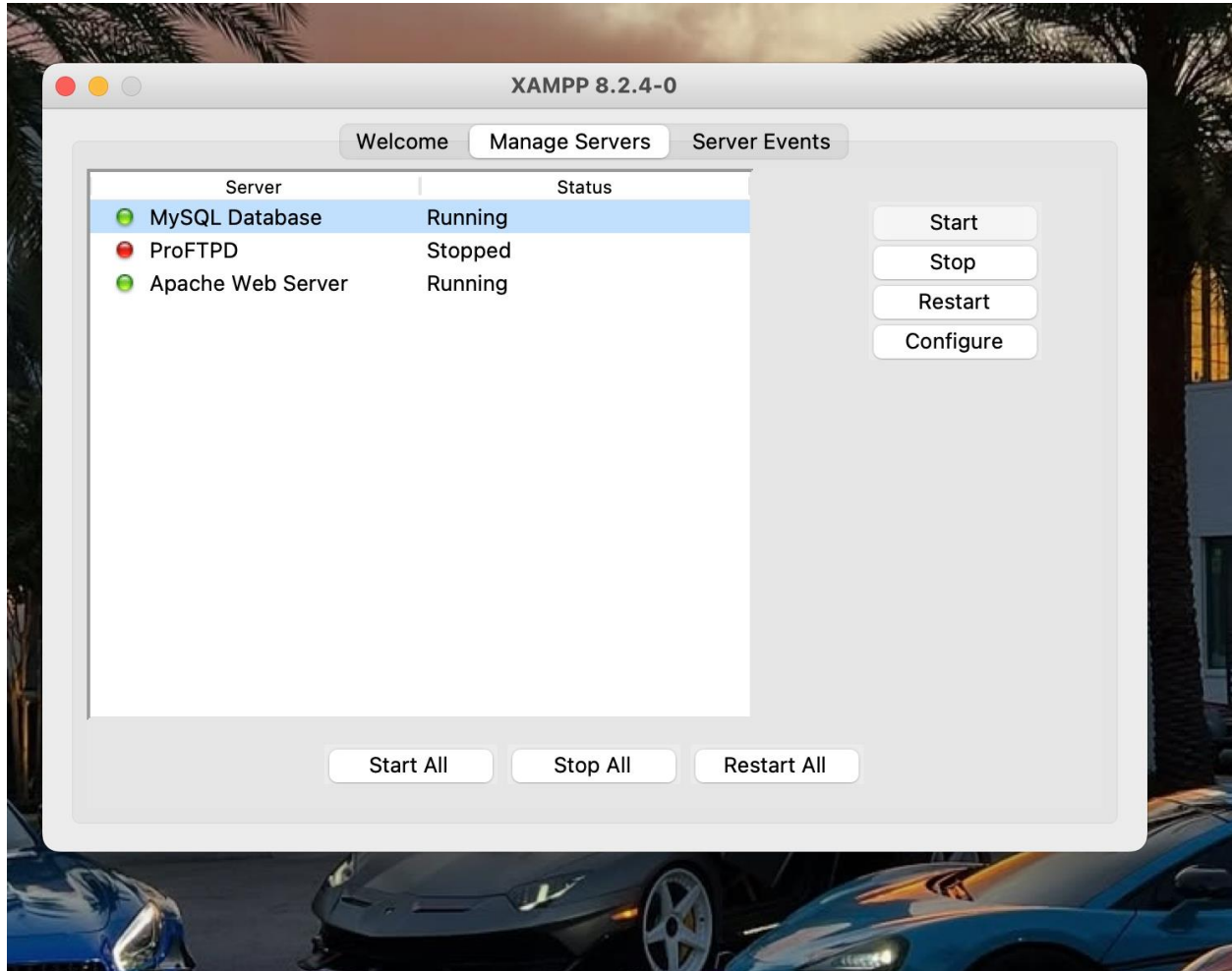


Fig 3; Local host test page confirming Apache serves content



Welcome to XAMPP for OS X 8.2.4

You have successfully installed XAMPP on this system! Now you can start using Apache, MariaDB, PHP and other components. You can find more info in the [FAQs](#) section or check the [HOW-TO Guides](#) for getting started with PHP applications.

XAMPP is meant only for development purposes. It has certain configuration settings that make it easy to develop locally but that are insecure if you want to have your installation accessible to others.

Start the XAMPP Control Panel to check the server status.

Community

XAMPP has been around for more than 10 years – there is a huge community behind it. You can get involved by joining our [Forums](#), liking us on [Facebook](#), or following our exploits on [Twitter](#).

Fig 4; First PHP page execution—Hello world test from htdocs



Fig 5: Database connection test using mysqli_connect[]



MY REFLECTION:

In Week 1, I installed XAMPP on macOS and resolved the security block via System Settings > Privacy & Security. I started Apache and MySQL from the control panel. The localhost test (Fig 3) confirmed the server was running. I created a project folder inside htdocs and wrote a simple PHP script (Fig 4). The database connection script (Fig 5) successfully connected to MySQL using `mysqli_connect()`. A challenge was understanding that files must be inside htdocs to be accessible via localhost. Now my local environment is ready for dynamic web development.

WEEK 2: WIREFRAMES AND GUI DESIGN

TITLE: USER INTERFACE PLANNING AND SYSTEM DESIGN

Fig 1: Hand drawn wireframe of the login page showing email, password fields and action buttons.

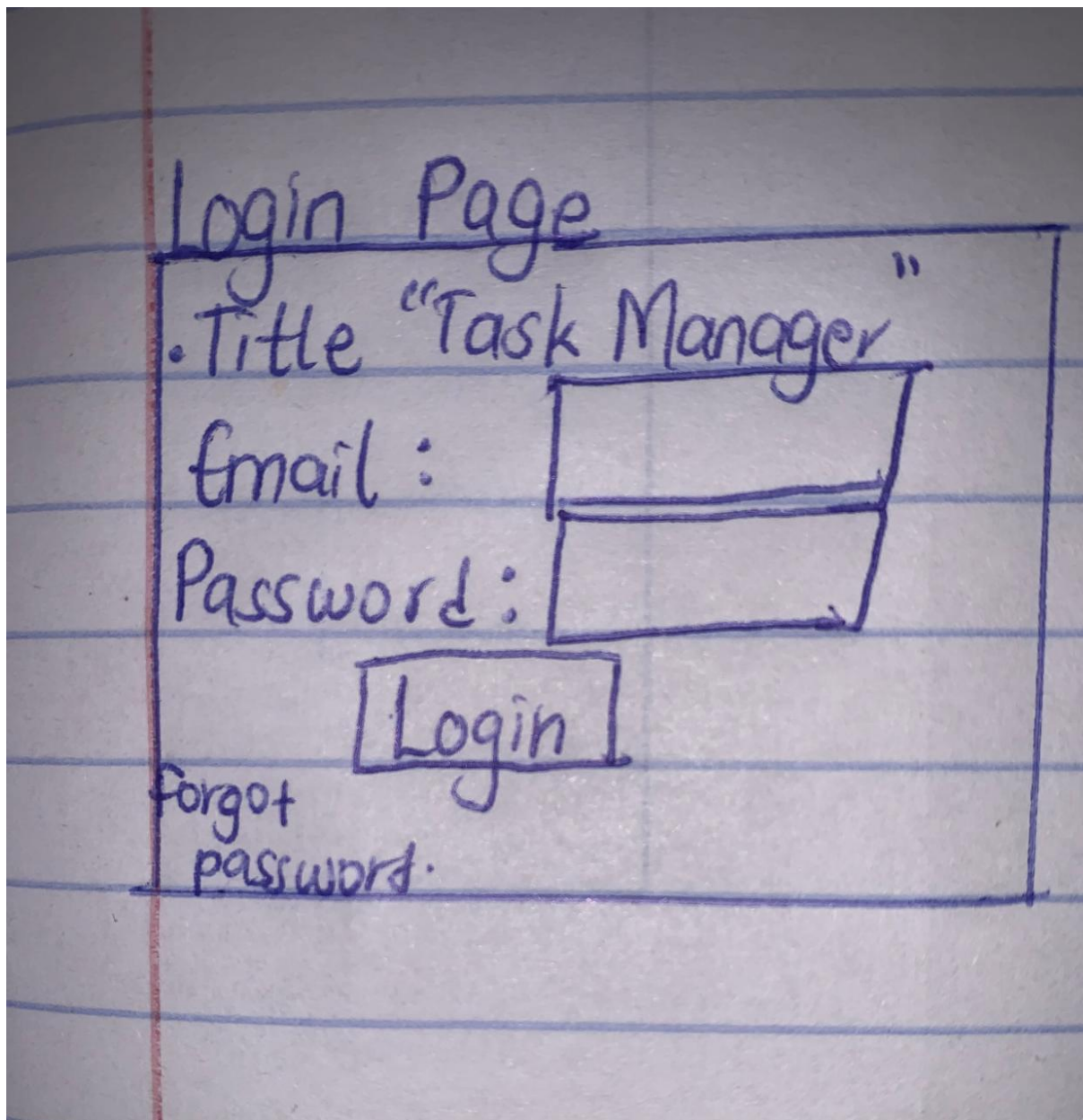


FIG 2; Hand drawn Dashboard layout with left sidebar navigation and main content area containing summary cards and task table

Dashboard:

Home	Task	Due	Status	Action
Add task	Name	Date		
Settings				
Logout				

FIG 3: Mobile Mockup showing stacked card layout and hamburger navigation for small screens

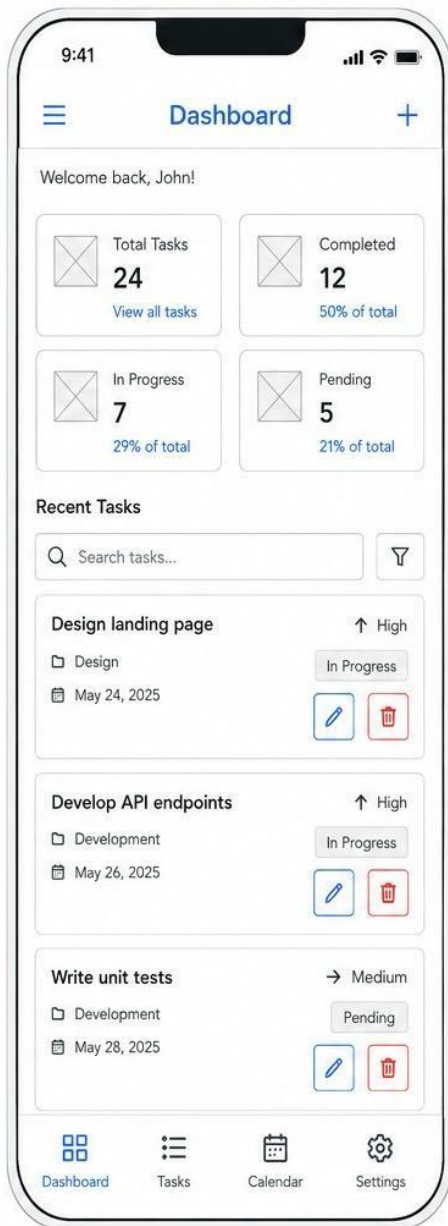


FIG 4: Colour theme and Navigation structure for desktop and mobile

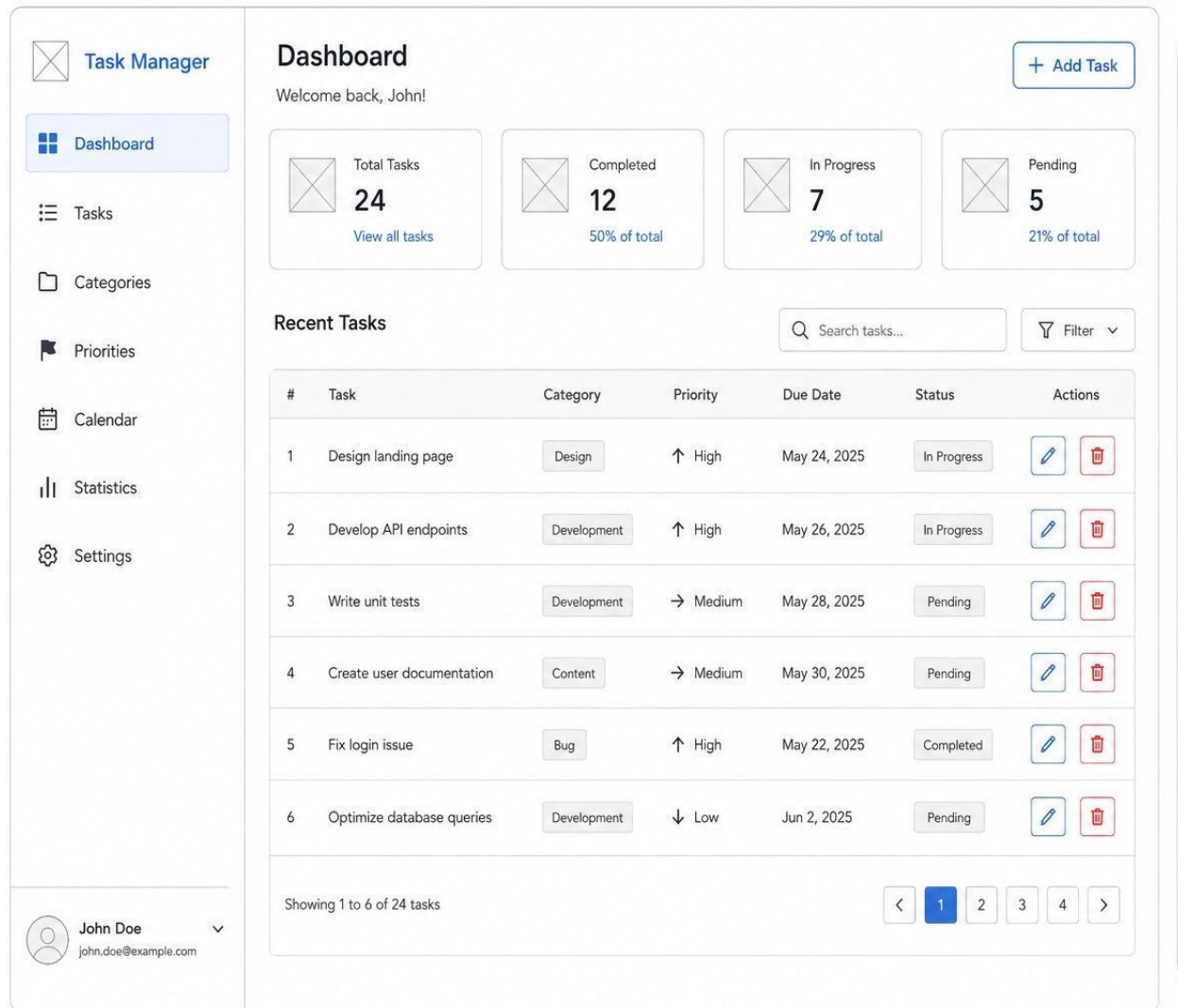
- Primary colour : #3498db (blue) for buttons.
- Secondary : #2c3e50 (dark blue) for sidebar
- Background : #ecf0f1 (light grey)
- Navigation : Always visible sidebar on desktop
hidden behind hamburger on mobile

FIG 5: Digital prototype created in canva showing the login page and dashboard layout

The image shows a digital prototype of a login page and dashboard layout. The login page is centered on a light gray background. It features a white rounded rectangle containing the following elements:

- Welcome Back!**: A bold heading at the top.
- Sign in to your account to continue**: A subtitle below the heading.
- : A placeholder for a profile picture.
- Email Address**: A label for the email input field.
- : An email input field with an envelope icon on the left.
- Password**: A label for the password input field.
- : A password input field with a lock icon on the left and an eye icon on the right to toggle visibility.
- ☐ **Remember me**: A checkbox and label.
- [Forgot password?](#): A link to the forgot password page.
- Login**: A blue button with white text.
- or**: A small circle with the word "or" inside, centered between two horizontal lines.
-  **Continue with Google**: A button with the Google logo and text.
- Don't have an account? [Sign up](#)**: A link to the sign up page.

The dashboard layout is partially visible at the bottom of the image, showing a light blue header bar.



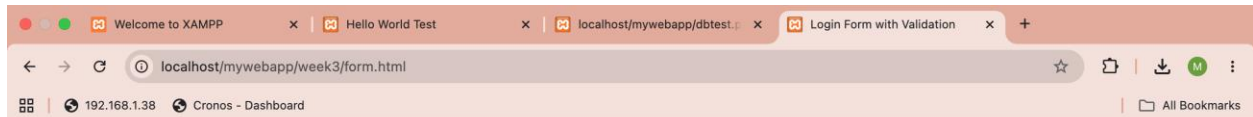
MY REFLECTION;

For Week 2, I designed the user interface for a Task Manager. I chose a left sidebar because it allows quick navigation and remains visible on larger screens. On mobile, the sidebar collapses into a hamburger menu to save space. The colour theme (dark blue-grey with teal accents) was selected for readability and a professional look. Hand-drawn wireframes helped me iterate layout ideas before moving to Canva for the digital prototype. A challenge was balancing information density on mobile – solved by stacking cards vertically and simplifying the task table.

WEEK 3; JAVASCRIPT AND PHP BASICS

TITLE : FRONTEND INTERACTION AND BACKEND FOUNDATIONS

Fig 1: JavaScript form validation preventing empty email/password submission



Login Form

Email:

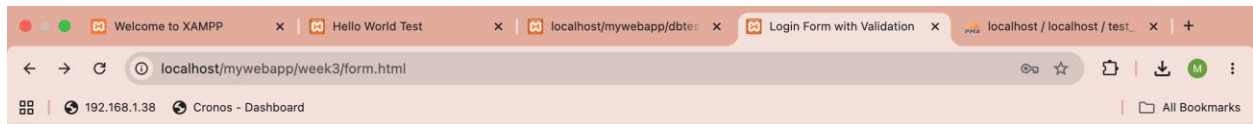
Email is required

Password:

Password is required

Login

FIG 2: PASSWORD STRENGTH CHECKER GIVING REAL TIME FEEDBACK

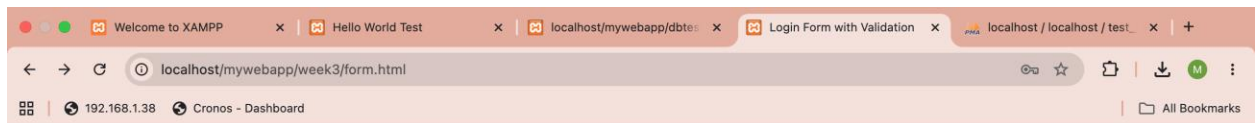


Login Form

Email:

Password:

Weak

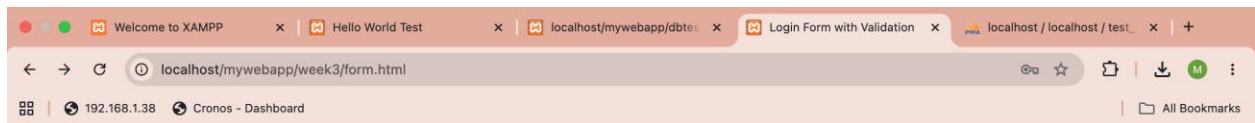


Login Form

Email:

Password:

Medium



Login Form

Email:

Password:

Strong

FIG 3: PHP processing of form data- shows email and password length



FIG 4: Database connection script showing PHP can connect to MySQL

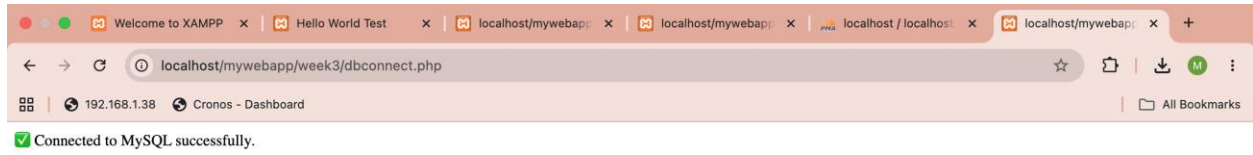
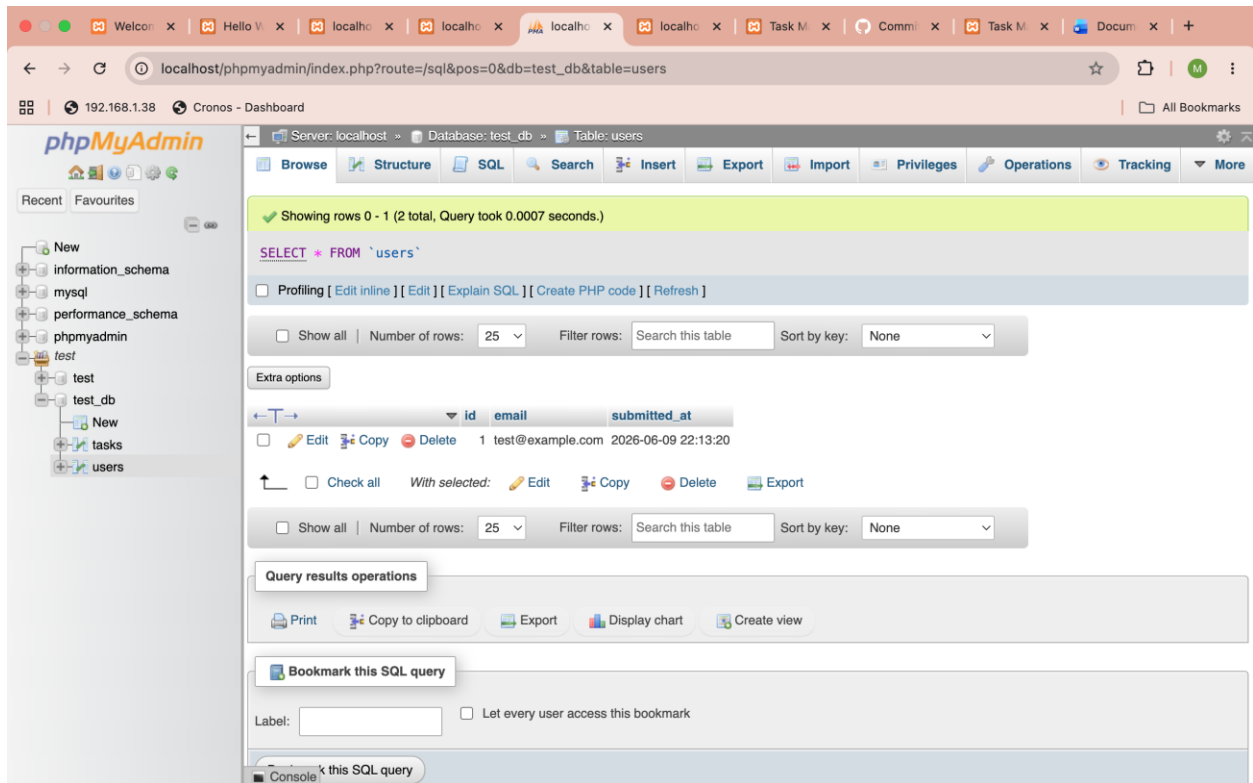


FIG 5 : DYNAMIC USER INPUT HANDLING- EMAIL STORED IN MYSQL DATABASE



MY REFLECTION :

In Week 3, I implemented a login form with JavaScript validation (Fig 1) that prevents empty submissions and checks for @ in email. A password strength meter (Fig 2) gives real-time feedback. The PHP script (process.php) receives and displays the submitted data (Fig 3). I confirmed the database connection (Fig 4) and stored the email into a MySQL table (Fig 5). A challenge was ensuring the form action path pointed to the correct PHP file. This week demonstrated how HTML, JavaScript, PHP, and MySQL work together in a complete flow.

WEEK 4: CRUD OPERATIONS {Create, Read, Update, Delete}

TITLE:FULL DATA MANAGEMENT WITH TASK MANAGER

FIG 1: Dashbard displaying all tasks from the database[READ]operation

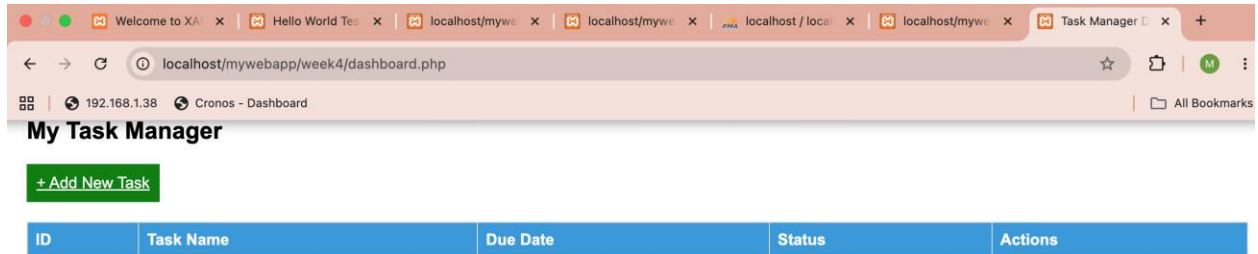


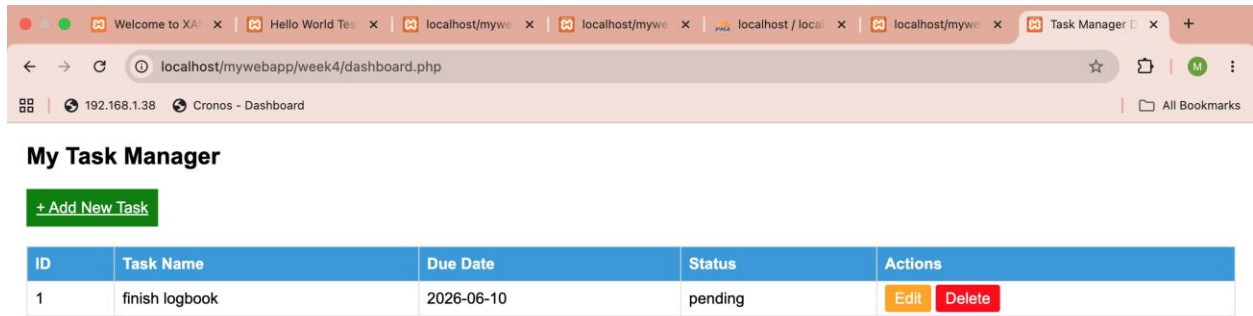
Fig 2: Form for creating a new task {create} operation

The screenshot shows a web browser window with multiple tabs. The active tab is titled "Add New Task" and the address bar shows the URL "localhost/mywebapp/week4/add_task.html". The browser's status bar at the bottom indicates the IP address "192.168.1.38" and the page title "Cronos - Dashboard".

The form itself is titled "Add New Task" and contains the following elements:

- Task Name:** A text input field containing the value "finish logbook".
- Due Date:** A date input field showing "10/06/2026" with a calendar icon to its right.
- Status:** A dropdown menu currently displaying "Pending".
- Save Task:** A green button with white text.
- Back to Dashboard:** A purple text link with an arrow icon.

Fig 3: Task Inserted into database with confirmation message



The screenshot shows a web browser window with multiple tabs. The active tab is titled "Task Manager" and displays the URL "localhost/mywebapp/week4/dashboard.php". Below the browser window, the page content is visible, featuring a heading "My Task Manager" and a green button labeled "+ Add New Task". A table lists tasks, with the first task having ID 1, Task Name "finish logbook", Due Date "2026-06-10", and Status "pending". The Actions column for this task contains "Edit" and "Delete" buttons.

My Task Manager

[+ Add New Task](#)

ID	Task Name	Due Date	Status	Actions
1	finish logbook	2026-06-10	pending	Edit Delete

FIG 4: Updating an Existing task

Edit Task

Task Name:

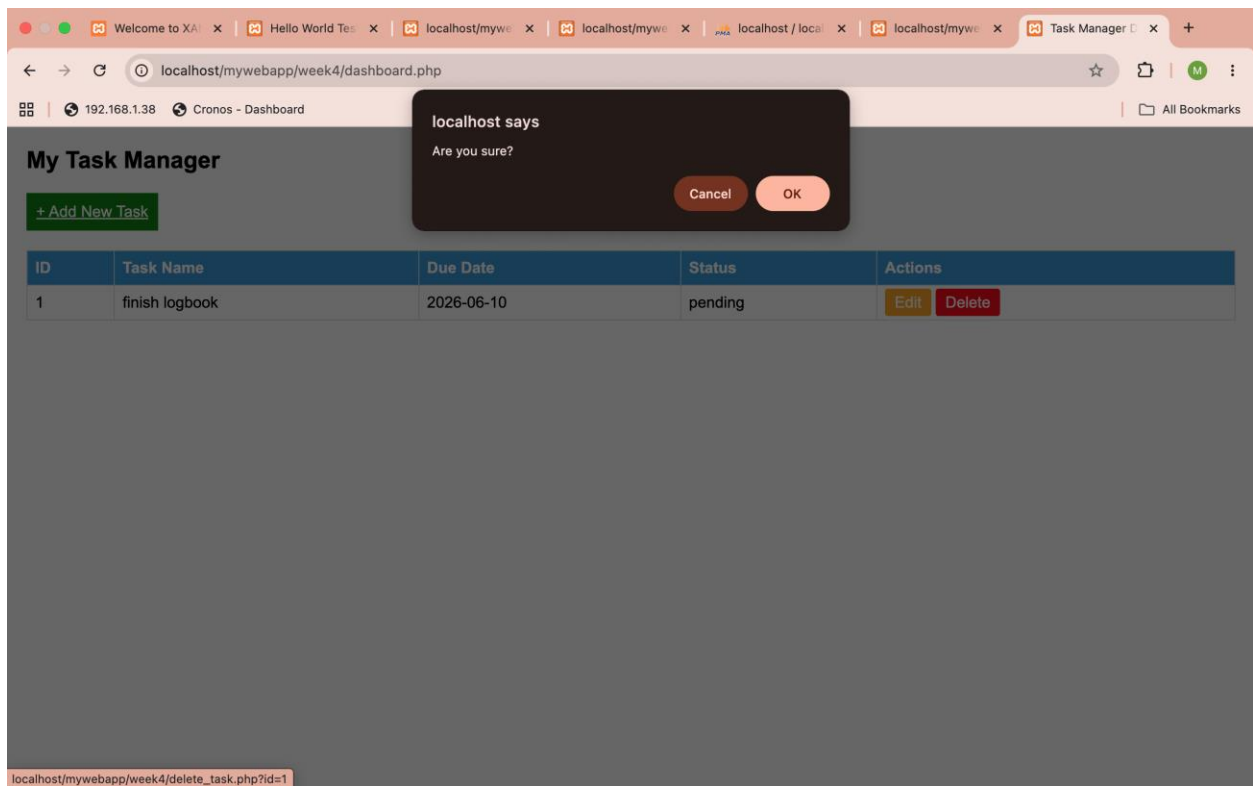
Due Date:

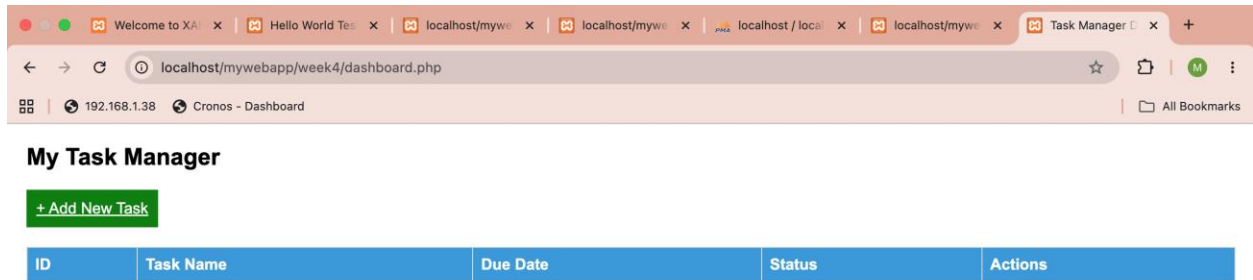
Status:

[← Back to Dashboard](#)

[Update Task](#)

Fig 5: Delete operation—Task removed from database after confirmation





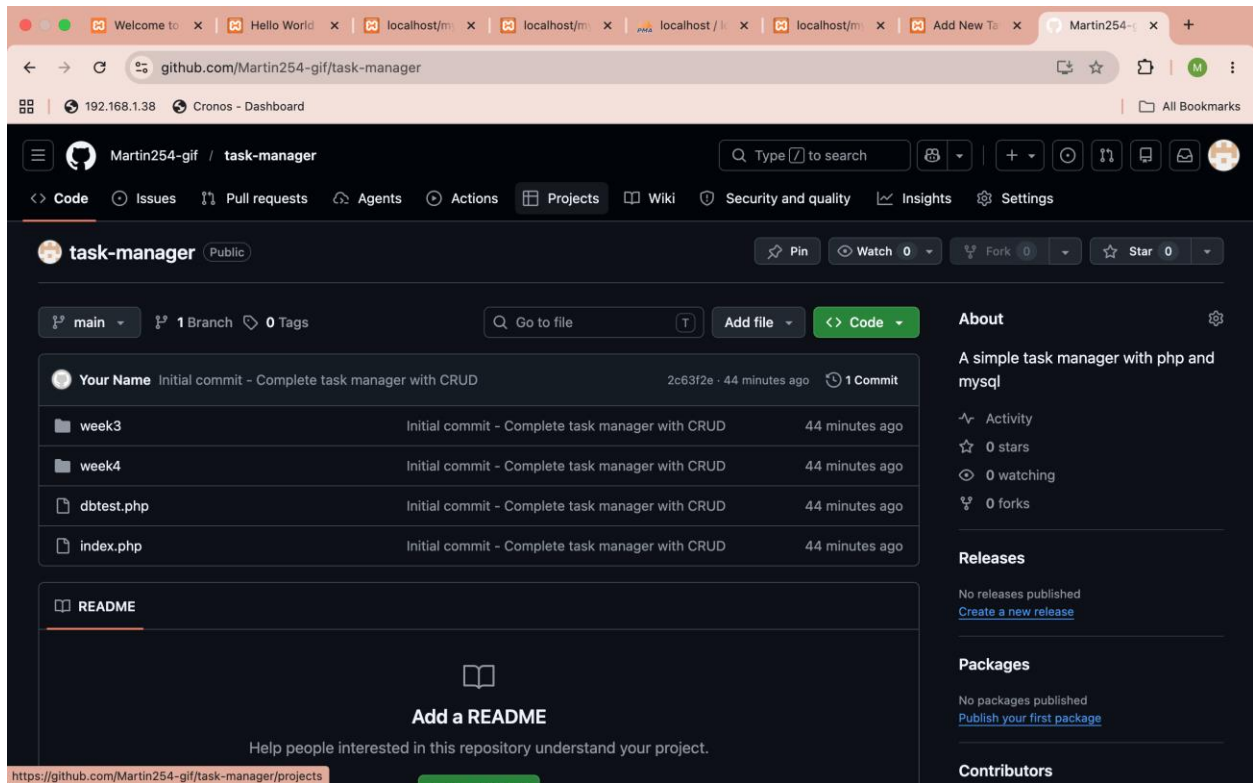
MY REFLECTION : In Week 4, I implemented full CRUD functionality for a Task Manager. The dashboard (Fig 1) reads and displays all tasks from the database. The add task form (Fig 2) inserts new tasks into the tasks table (Fig 3). The edit feature (Fig 4) updates existing records, and delete (Fig 5) removes them after a confirmation prompt. A challenge was passing the task ID via URL parameters to edit and delete pages. I learned that CRUD operations form the core of most web applications. Now my system can create, read, update, and delete tasks – a complete data management solution.

WEEK 5: VERSION CONTROL AND PORTFOLIO

SUBMISSION

TITLE: GITHUB INTEGRATION AND PROJECT COMPLETION

FIG 1: GITHUB repository containing complete project source code



**FIG 2: GIT COMMIT HISTORY DEMONSTRATING VERSION CONTROL
PROGRESS**

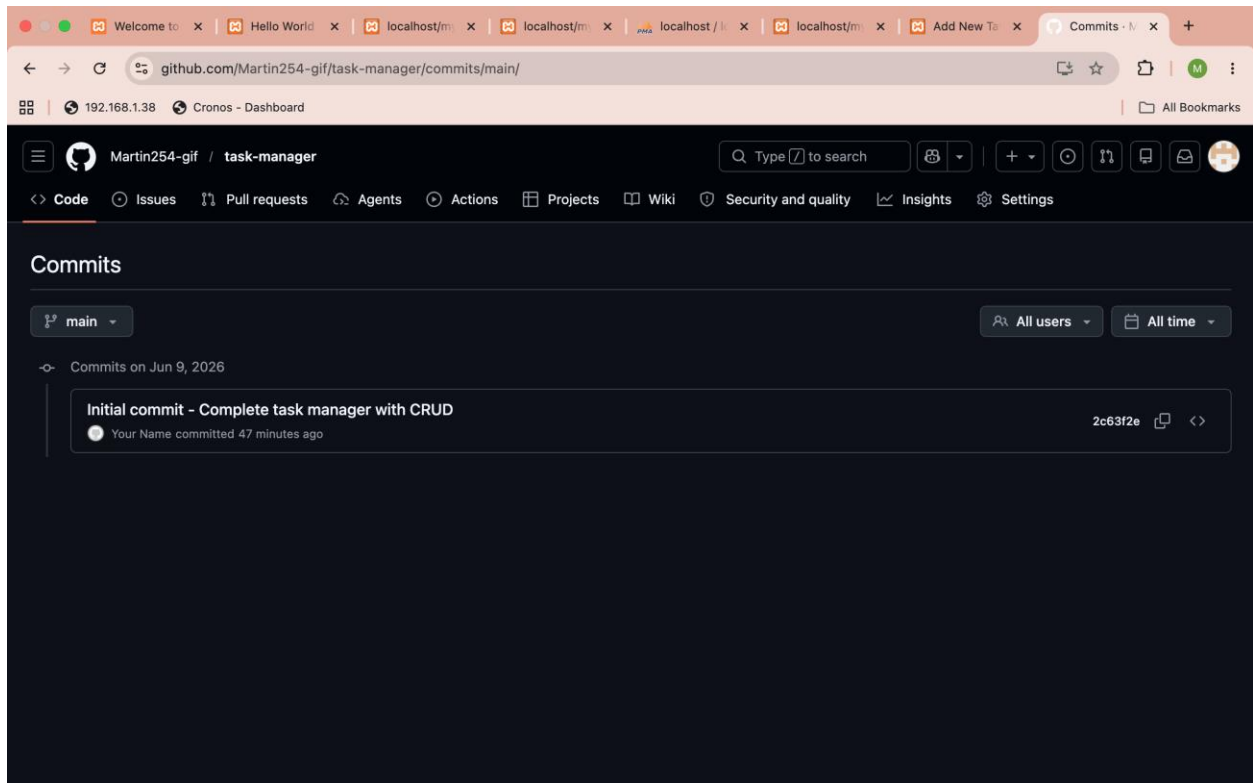


FIG 3: TASK MANAGER RUNNING ON LOCAL HOST—ALL CRUD OPERATIONS FUNCTIONAL

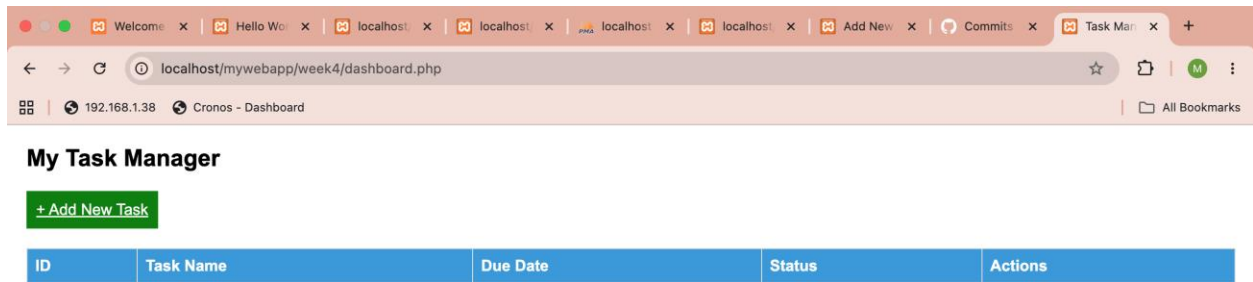


FIG 4: LOCAL PROJECT STRUCTURE WITH WEEKS 1-4 COMPLETED

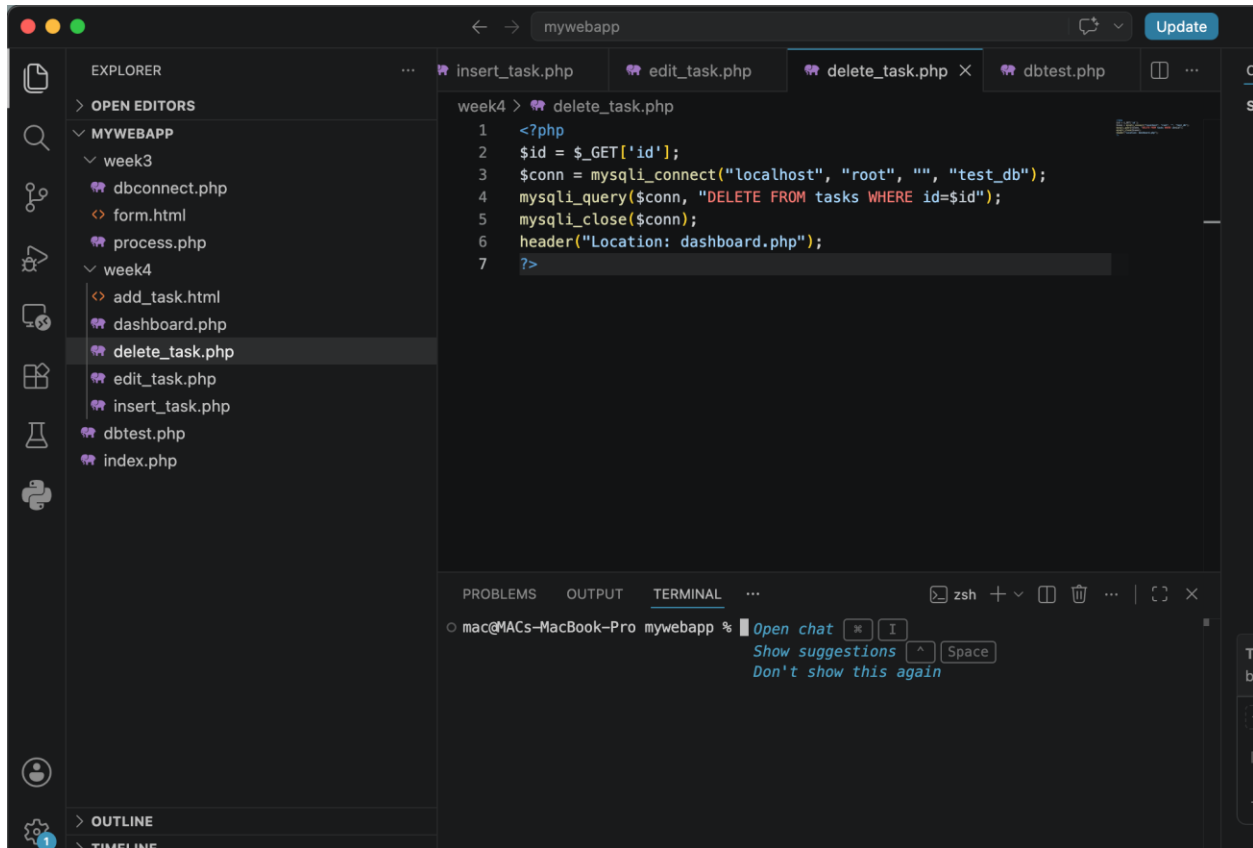
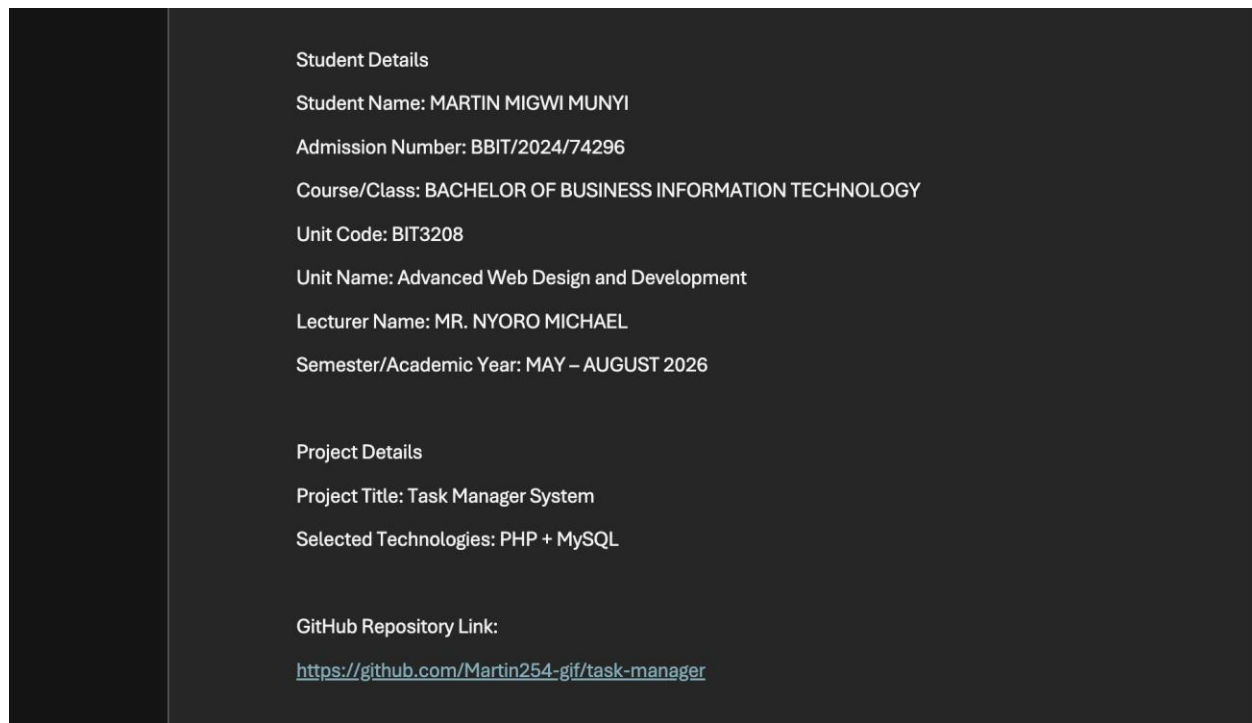


FIG 5: GITHUB REPOSITORY LINK INCLUDED ON THE COVER PAGE AS REQUIRED



MY REFLECTION:

In Week 5, I completed version control by pushing my entire project to a public GitHub repository (Fig 1). The commit history (Fig 2) shows my initial commit containing all CRUD operations. My local Task Manager runs correctly (Fig 3). The project structure (Fig 4) includes weeks 1 through 4. I have included the GitHub link on my logbook cover page (Fig 5). This fulfills the assignment requirement because students must provide either a GitHub repository, an online portfolio, or a live hosting link. I chose GitHub as my portfolio evidence. This also meets the industry expectation for version control and professional documentation.